SCIENCE 1.3 **Name:_____**

WORKSHEET SIX **INHERITANCE**

1. Match up the following terms with their appropriate descriptors. (A)

(a) Genotype		(1) A gene that has identical alleles
(b) Zygote		(2) The pattern of genes present in cells
(c) Phenotype		(3) An allele whose influence is more powerful than its partner
(d) Recessive allele		(4) A pair of alleles that carry different forms of the same gene
(e) Homozygous		(5) A sex cell
(f) Heterozygous		(6) A allele that is expressed only if the dominant form is absent
(g) Gamete		(7) First cell of the new organism
(h) Dominant allele		(8) The actual appearance produced by an organism's genes

2. Complete the paragraph using the words in the key list below. (A)

shape, variation, body height, symmetrical, population, Normal, Bell.

Many characteristics are found to display **continuous** _____. Examples would be body mass and _____. When a continuous variation pattern is graphed, the graph always displays the same _____. This is called a _____distribution curve or a _____ curve. However if the data for the graph is taken from too small a _____the graph may not have a _____ shape.

3. Read the following genetic "rules" for monohybrid crosses. Beside each rule put **T** or **F** depending on whether you think it is true or false. (A)

(a) Traits are controlled by a pair of genes.

(b) Recessive alleles mask dominant ones.

(c) Gametes have one allele for each trait.

(d) Different forms of a gene for the same trait are called alleles.

(e) The genotype describes the appearance.

(f) The phenotype lists the alleles present.

(g) In heterozygous individuals one allele is dominant and one is recessive.

(h) An individual displaying the dominant trait could have one of two genotypes.

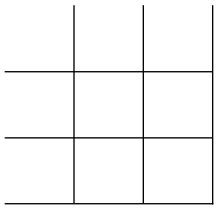
4. In guinea-pigs, a black colour coat (B) is dominant to a brown colour coat (b).

(a) Write the genotype of a homozygous dominant individual. (A) _____

(b) Write the genotype of a heterozygous individual. (A) _____

(c) Write the genotype of a homozygous recessive individual. (A) _____

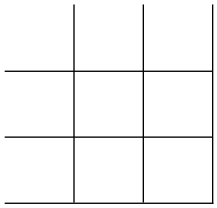
(d) A heterozygous black male was mated with a homozygous brown female. Complete the punnet square to show the possible offspring genotypes. (A)



(e) What ratio of brown to black offspring is suggested by the the completed punnet square. (M) _____

5. Humans either have the ability to roll their tongue (T) or they do not (t). Tongue rolling is dominant over non-tongue rolling. A married couple who are both tongue rollers have three children. One child is a non-tongue roller and the other two are tongue rollers.

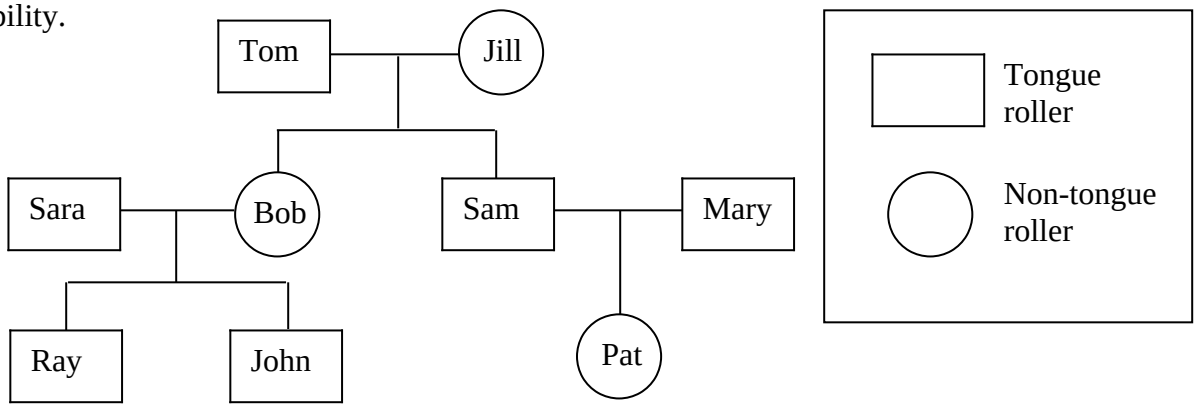
(a) Draw a punnet square that shows how this could happen.(M)



(b) Is it possible for one of the children to be heterozygous? _____ . Write this child's genotype. _____. (A)

(c) What is the phenotype ratio derived from this punnet square? (M) _____

6. An inheritance chart is shown below for three generations of a family regarding tongue-rolling ability.



Using the chart to help you, answer the following questions. (M)

(a) Write genotypes for Tom _____ and for Jill _____.

(b) How did you know whether Tom was Tt or TT? _____

_____

(c) Write genotypes for Sam _____ and Mary _____ and John _____.

(b) How did you know whether Mary was Tt or TT? _____

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